

McStas - Mantid

- and introducing McStasToX
for use with Scipp

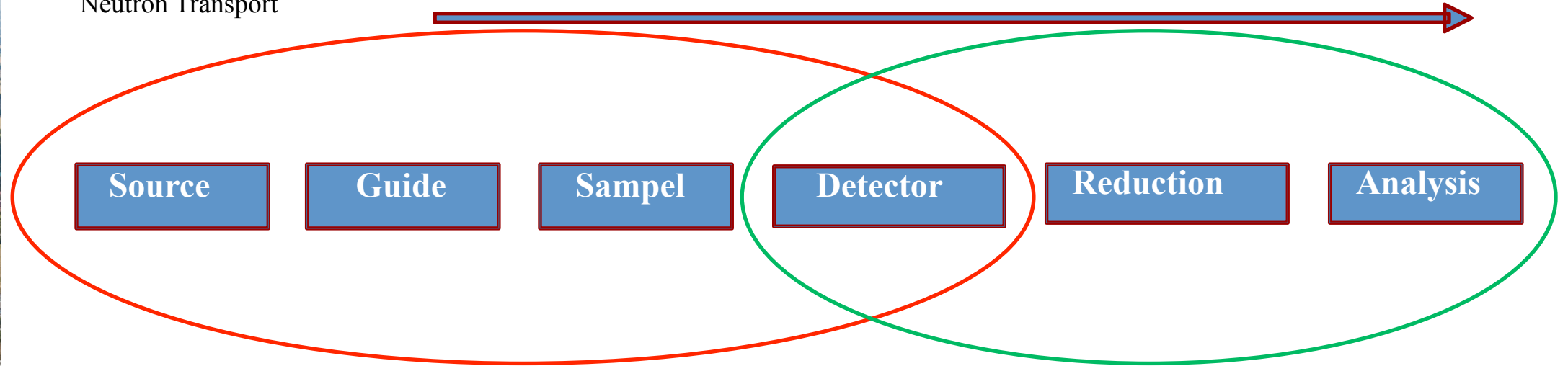
Torben Nielsen

Anders Markvardsen

Peter Willendrup



Neutron Transport



Typical McStas Neuron Transport

Mantid-McStas focus:

How to analyse McStas data

2025 ISIS
McStas
School



Comparison

Classic McStas

Post processing:
iFit/ Matlab/ own code



mcstas - plot



mcstas - transport

Mantid-McStas

Post processing:
Mantid algorithms



mantidplot



mcstas - transport

2025 ISIS
McStas
School

Motivation: new flexible dataflow



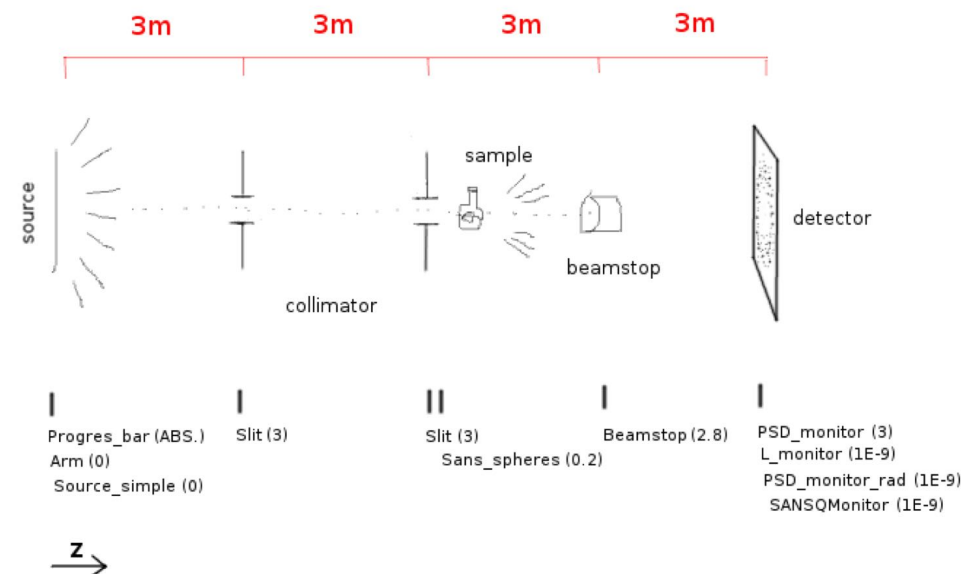
1. Combine the reduction framework and the neutron transport framework
2. Neutron transport, reduction and “analysis” in one go
3. View McStas data in Mantid (histogram data)
4. Import McStas event data to Mantid
5. Use already developed and tested algorithms in Mantid to process McStas event data (no need to reinvent methods)
6. Advanced use: Use McStas to quantify (perhaps remove) “spurries” signal on detector (can scatt. –multi. scatt)

Illustration: simplified SANS

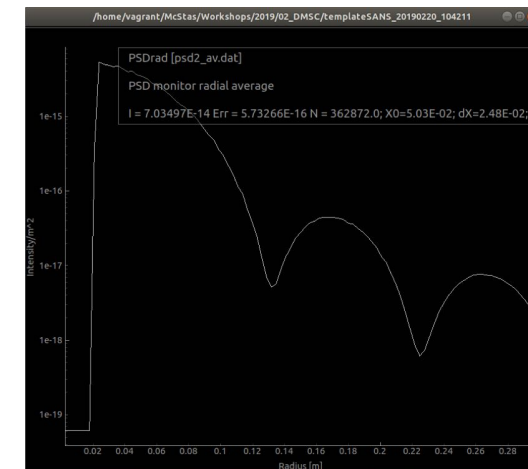
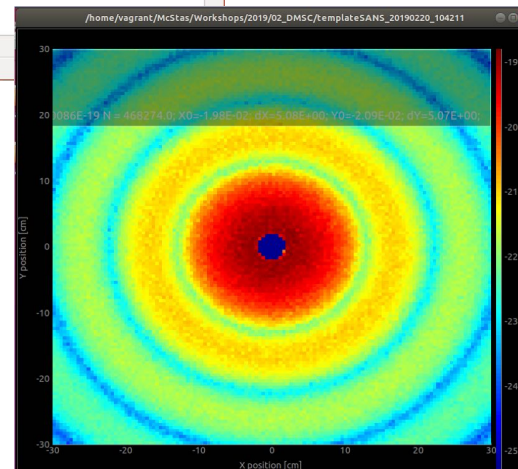
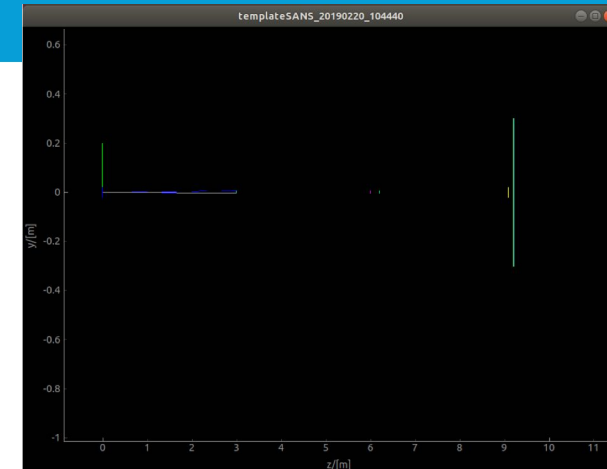
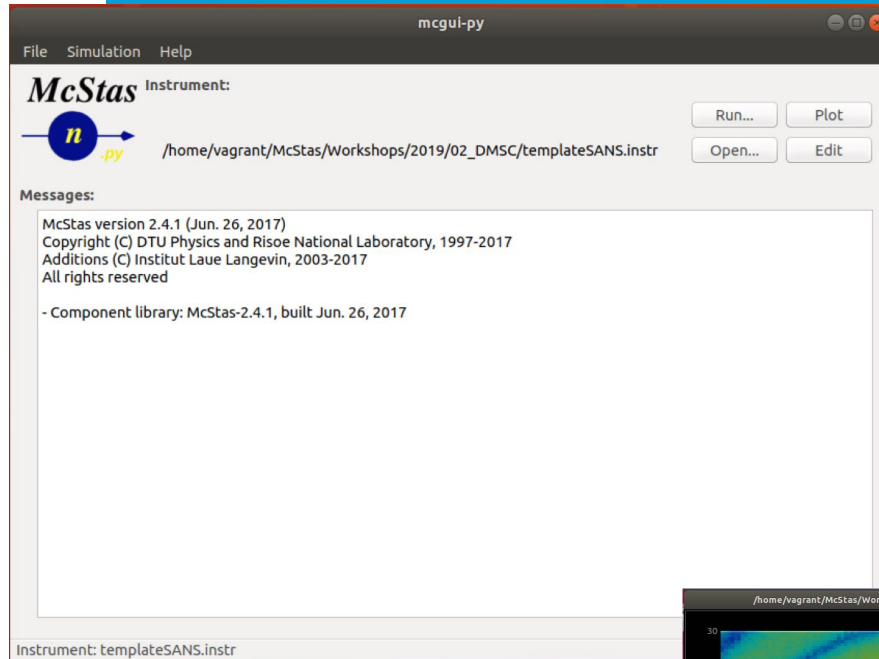


In this tutorial, you will write a simplified SANS instrument. When you have completed this tutorial, you will have learned the basics of mcstas.

- Requirements: mcstas 2.2a.

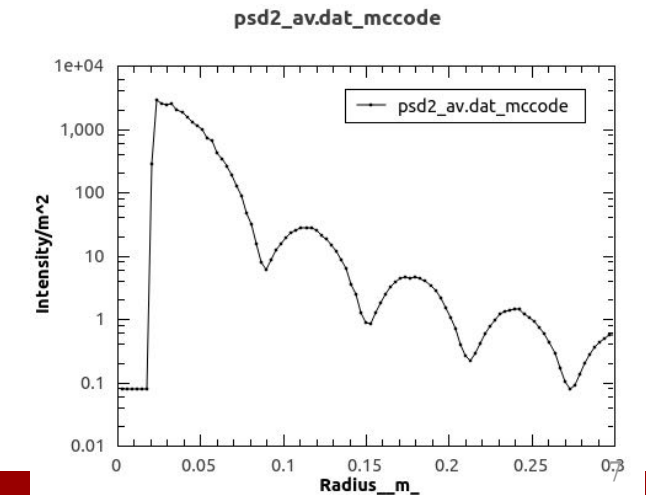
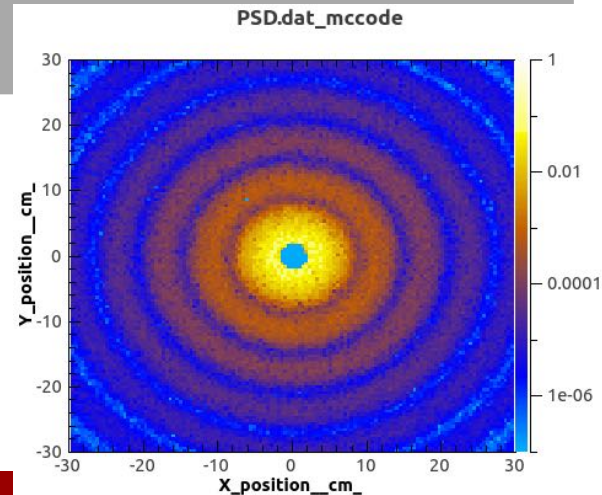
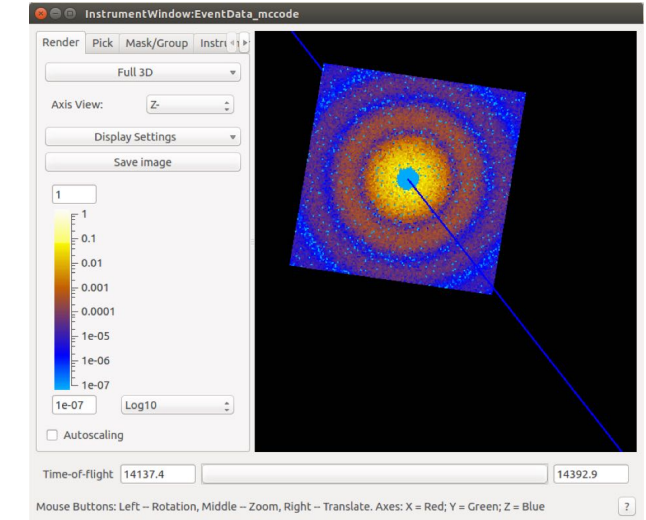
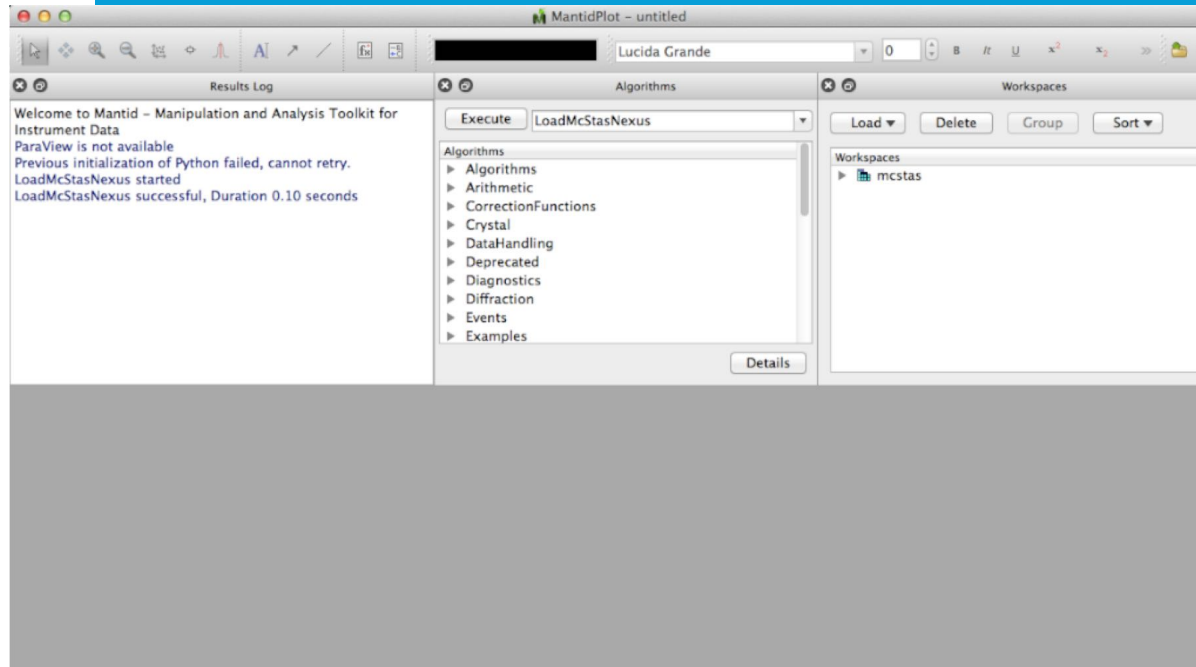


McStas GUI:

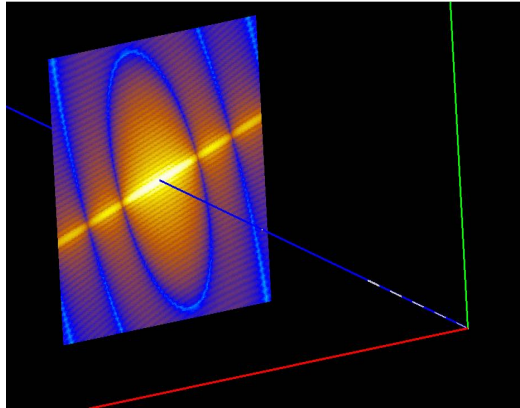


$R=150, \phi=0.1, d=1, a=0$

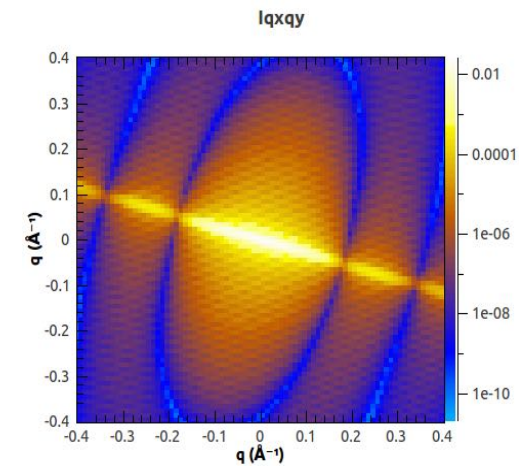
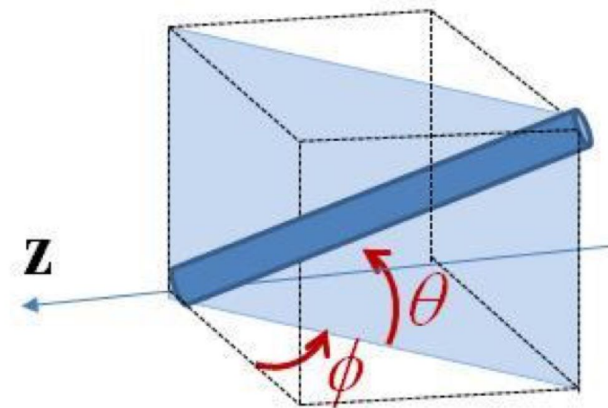
Mantid GUI:



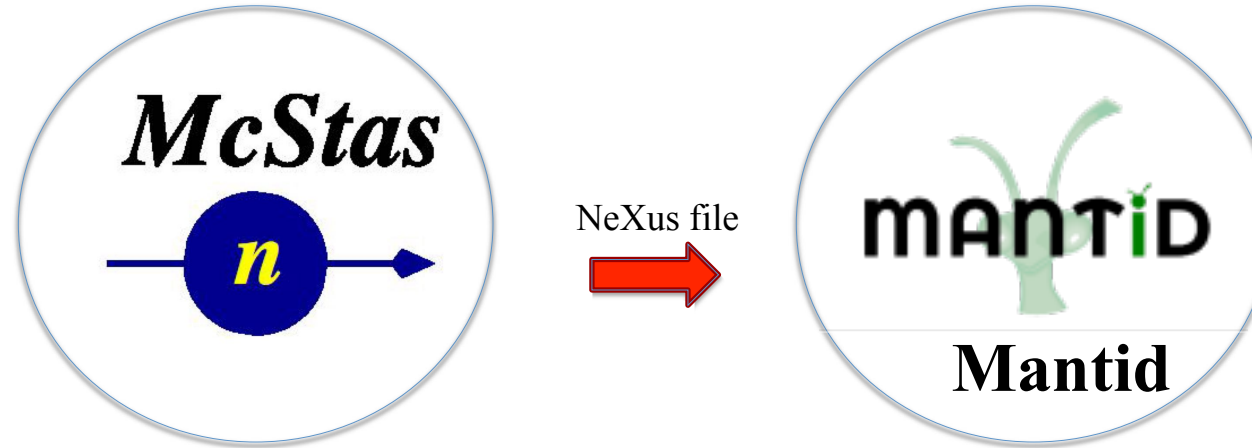
Example: Usage of Mantid



- McStas –Mantid
- 2D scattering kernel from SasView
- 2D reduction in Mantid: Qxy
- Can be send back to SasViewfitting



The NeXus file



- The McStas Nexus file must contain:
- Event data, i.e. each neutron has a pixel id and a time stamp
- An IDF McStas monitor_nD gives pixel-ID & time for each event
- *mcdisplay-mantid can auto-generate an IDF*

McStas NeXus file

HDFView 2.9

File Window Tools Help

Recent Files: /home/ub/tmp/McStas_Event/McStas/Exp_Data/mccode.h5

TableView - events - /entry1/data/k01_events_dat_list_p_x_y_n_id_t/ - /home/ub/tmp/McStas_Event/McStas/Exp_Data/mccode.h5

Table

	0	1	2	3	4	5
0	3.185987...	-0.01450...	0.007932...	0.0	18717.0	0.025524...
1	0.192848...	0.141796...	0.347311...	1.0	31227.0	0.046068...
2	2.989055...	-0.01409...	-5.33756...	2.0	18333.0	0.025703...
3	0.001396...	-0.00520...	-0.01711...	3.0	17759.0	0.025971...
4	3.701651...	0.251736...	-0.12745...	4.0	13776.0	0.008363...
5	0.032482...	-0.11949...	0.083006...	5.0	21385.0	0.018234...
6	2.859783...	0.004956...	0.005223...	6.0	18720.0	0.041336...
7	0.022012...	0.058801...	-0.08114...	7.0	15720.0	0.041336...
8	8.530934...	0.012643...	-0.03070...	8.0	17720.0	0.041336...
9	6.984646...	-0.00867...	0.012712...	9.0	18720.0	0.041336...
10	0.009140...	-0.07853...	-0.07291...	10.0	15720.0	0.041336...
11	7.150334...	-0.01082...	0.005151...	11.0	18720.0	0.041336...
12	0.001010...	0.457856...	0.087702...	12.0	21720.0	0.041336...
13	0.004965...	0.003146...	-0.04262...	13.0	16720.0	0.041336...
14	8.629457...	-0.03193...	-0.02392...	14.0	17720.0	0.041336...
15	7.072494...	0.015294...	-0.01158...	15.0	17720.0	0.041336...
16	3.898581...	-2.38144...	0.011657...	16.0	18720.0	0.041336...
17	3.479036...	-0.02370...	-0.00281...	17.0	18720.0	0.041336...
18	5.870513...	0.006430...	0.006773...	18.0	18720.0	0.041336...
19	0.010325...	-0.07164...	-0.01926...	19.0	17720.0	0.041336...
20	3.191063...	-0.00829...	-0.00252...	20.0	18720.0	0.041336...
21	3.746034...	-0.00570...	0.011053...	21.0	18720.0	0.041336...
22	0.003724...	0.038904...	0.029156...	22.0	19720.0	0.041336...
23	3.316878...	-0.00442...	6.097504...	23.0	18720.0	0.041336...
24	0.023456...	-0.03083...	0.084904...	24.0	21720.0	0.041336...
25	0.001505...	0.031248...	-0.02562...	25.0	17720.0	0.041336...

Event data

TextView - data - /entry1/instrument/instrument_xml/ - mccode.h5

Text

Data selection: [0] ~ [0]

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- IDF generated using McStas McDisplay and the Mantid backend -->
<!-- For help on the notation used to specify an Instrument Definition File see
http://www.mantidproject.org/IDF -->
<instrument name="ISIS_SANS2d_Mantid.out" valid-from="1900-01-31 23:59:59"
valid-to="2100-01-31 23:59:59" last-modified="Tue Apr 4 14:17:50 2017">
<defaults>
  <length unit="meter"/>
  <angle unit="degree"/>
  <reference-frame>
    <!-- The z-axis is set parallel to and in the direction of the beam. The
y-axis points up and the coordinate system is right handed. -->
    <along-beam axis="z"/>
    <pointing-up axis="y"/>
    <handedness val="right"/>
  </reference-frame>
  <default-view axis-view="z"/>
</defaults>
</instrument>
</xml>
```

IDF xml data

data (22370, 2)
String, length = 7891, 1
Number of attributes = 0

Log Info Metadata



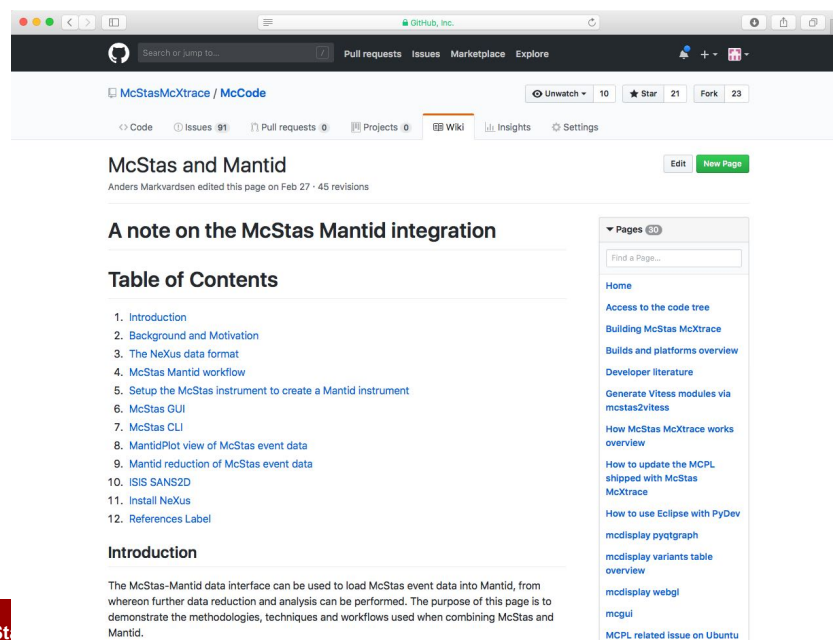
IDF xml data, TOF and pixel ID's

- Mantid's IDF store geometry information use in TOF analysis
- This implies parsing information about:
 - where the neutron source is located,
 - where the sample is located,
 - where each individual detector pixel is located.

In other words:
Establishing k_i and k_f

How to use: Online documentation

- Github McStas wiki pages
(<https://github.com/McStasMcXtrace/McCode/wiki/McStas-and-Mantid#setup-the-mcstas-instrument-to-create-a-mantid-instrument>)
- Archive - lanl.arXiv.org
- Built-into Mantid



McStas and Mantid
Anders Markvardsen edited this page on Feb 27 · 45 revisions

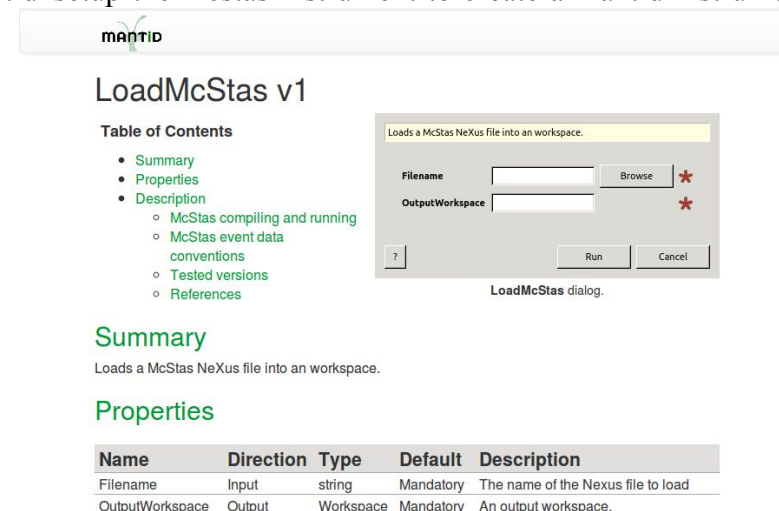
A note on the McStas Mantid integration

Table of Contents

1. Introduction
2. Background and Motivation
3. The NeXus data format
4. McStas Mantid workflow
5. Setup the McStas instrument to create a Mantid instrument
6. McStas GUI
7. McStas CLI
8. MantidPlot view of McStas event data
9. Mantid reduction of McStas event data
10. ISIS SANS2D
11. Install NeXus
12. References Label

Introduction

The McStas-Mantid data interface can be used to load McStas event data into Mantid, from whereon further data reduction and analysis can be performed. The purpose of this page is to demonstrate the methodologies, techniques and workflows used when combining McStas and Mantid.



LoadMcStas v1

Table of Contents

- Summary
- Properties
- Description
 - McStas compiling and running
 - McStas event data conventions
 - Tested versions
 - References

LoadMcStas dialog.

Loads a McStas NeXus file into an workspace.

Filename Browse 

OutputWorkspace 

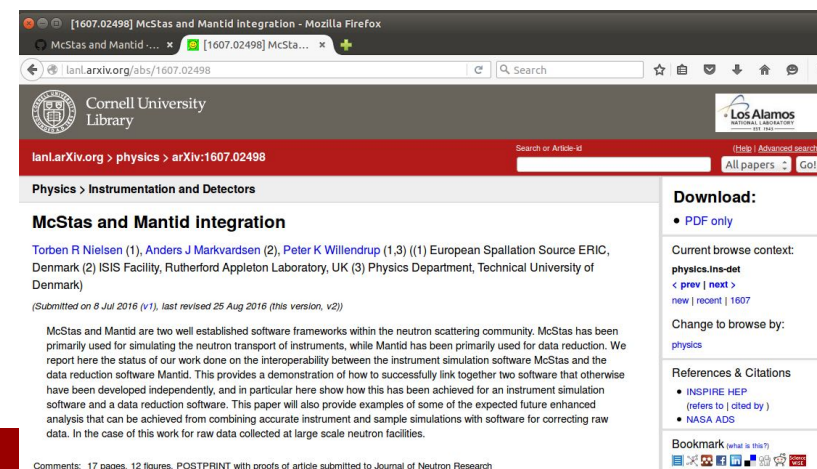
Run Cancel

Summary

Loads a McStas NeXus file into an workspace.

Properties

Name	Direction	Type	Default	Description
Filename	Input	string	Mandatory	The name of the Nexus file to load
OutputWorkspace	Output	Workspace	Mandatory	An output workspace.



McStas and Mantid Integration

Torben R Nielsen (1), Anders J Markvardsen (2), Peter K Willendrup (1,3) ((1) European Spallation Source ERIC, Denmark (2) ISIS Facility, Rutherford Appleton Laboratory, UK (3) Physics Department, Technical University of Denmark)

(Submitted on 8 Jul 2016 (v1), last revised 25 Aug 2016 (this version, v2))

McStas and Mantid are two well established software frameworks within the neutron scattering community. McStas has been primarily used for simulating the neutron transport of instruments, while Mantid has been primarily used for data reduction. We report here the status of our work done on the interoperability between the instrument simulation software McStas and the data reduction software Mantid. This provides a demonstration of how to successfully link together two software that otherwise have been developed independently, and in particular here show how this has been achieved for an instrument simulation software and a data reduction software. This paper will also provide examples of some of the expected future enhanced analysis that can be achieved from combining accurate instrument and sample simulations with software for correcting raw data. In the case of this work for raw data collected at large scale neutron facilities.

Comments: 17 pages, 12 figures, POSTPRINT with proofs of article submitted to Journal of Neutron Research
Subjects: Instrumentation and Detectors (physics.ins-det)



McStas instrument KEYWORDS

(meaning: set up definition of k_i and k_f)



McStas instrument file name and the McStas defined name of the instrument must be the same:

- E.g. **templateSANS_Mantid.instr** and “**DEFINE INSTRUMENT**
templateSANS_Mantid(...)”

In the McStas instrument file the source must be named “**sourceMantid**”

- E.g. “**COMPONENT sourceMantid= Source_simple(...)**”

In the McStas instrument file the sample must be named “**sampleMantid**”

- E.g. “**COMPONENT sampleMantid= Sans_spheres(...)**”

In the McStas instrument file the event monitors must be named “**nD_Mantid_#**”

- E.g. “**COMPONENT nD_Mantid_1 = Monitor_nD(...)**”

templateSANS.inst vs templateSANS_Mantid.inst



```

templateSANS.instr
UNREGISTERED
1  DEFINE INSTRUMENT templateSANS(lambda=6, dlambda=0.05, r=100, PHI=1e-
2  3, Delta_Rho=0.6, sigma_abs=0.5)
3  TRACE
4
5  COMPONENT a1 = Progress_bar()
6  AT (0,0,0) ABSOLUTE
7
8  COMPONENT arm = Arm()
9  AT (0, 0, 0) ABSOLUTE
10
11 COMPONENT source = Source_simple(
12     radius = 0.02, dist = 3, focus_xw = 0.01, focus_yh = 0.01,
13     lambda0 = lambda, dlambda = dlambda, flux = 1e16)
14 AT (0, 0, 0) RELATIVE arm
15
16 COMPONENT coll1 = Slit(
17     radius = 0.005)
18 AT (0, 0, 3) RELATIVE arm
19
20 COMPONENT coll2 = Slit(
21     radius = 0.005)
22 AT (0, 0, 6) RELATIVE arm
23
24
25
26
27
28
29
30 SPLIT COMPONENT sample = Sans_spheres(
31     R=r, Phi=PHI, Delta_rho=Delta_Rho, sigma_abs=sigma_abs,
32     xwidth=0.01, yheight=0.01, zdepth=0.005, focus_xw=0.6, focus_yh=0.
33     6, target_index=2)
34 AT (0,0,0.2) RELATIVE coll2
35
36 COMPONENT STOP = Beamstop(
37     radius = 0.02)
38 AT (0, 0, 2.9) RELATIVE sample

```

Line 1, Column 1 Spaces: 2 Plain Text

```

templateSANS_Mantid.instr
UNREGISTERED
1  DEFINE INSTRUMENT templateSANS_Mantid(lambda=6, dlambda=0.05, r=150,
2  PHI=1e-3, Delta_Rho=0.6, sigma_abs=0.0)
3  TRACE
4
5  COMPONENT a1 = Progress_bar()
6  AT (0,0,0) ABSOLUTE
7
8  COMPONENT arm = Arm()
9  AT (0, 0, 0) ABSOLUTE
10
11 COMPONENT sourceMantid = Source_simple(
12     radius = 0.02, dist = 3, focus_xw = 0.01, focus_yh = 0.01,
13     lambda0 = lambda, dlambda = dlambda, flux = 1e16)
14 AT (0, 0, 0) RELATIVE arm
15
16 COMPONENT coll1 = Slit(
17     radius = 0.005)
18 AT (0, 0, 3) RELATIVE arm
19
20 COMPONENT coll2 = Slit(
21     radius = 0.005)
22 AT (0, 0, 6) RELATIVE arm
23
24 COMPONENT ldetectorPRE = L_monitor(
25     nL = 1000, filename = "Edet0.dat", xmin = -0.3,
26     xmax = 0.3, ymin = -0.3, ymax = 0.3, Lmin = 5.5,
27     Lmax = 6.5)
28 AT (0,0,0.05) RELATIVE coll2
29
30 SPLIT COMPONENT sampleMantid = Sans_spheres(
31     R=r, Phi=PHI, Delta_rho=Delta_Rho, sigma_abs=sigma_abs,
32     xwidth=0.01, yheight=0.01, zdepth=0.005)
33 AT (0,0,0.2) RELATIVE coll2
34 EXTEND %{
35     if (!SCATTERED) ABSORB;
36     %}

```

Line 1, Column 1 Spaces: 2 Plain Text

templateSANS.inst vs templateSANS_Mantid.inst



2025 ISIS
McStas
School

```
COMPONENT detector = PSD_monitor( UNREGISTERED
COMPONENT detector = PSD_monitor(
COMPONENT nD_Mantid_1 = Monitor_nD(
1 COMPONENT detector = PSD_monitor(
2   nx = 128, ny = 128, filename = "PSD.dat", xmin = -0.3,
3   xmax = 0.3, ymin = -0.3, ymax = 0.3)
4   AT (0, 0, 3) RELATIVE sample
5
```

Line 5, Column 1 Tab Size: 4 Plain Text

```
COMPONENT nD_Mantid_1 = Monitor_nD( UNREGISTERED
COMPONENT detector = PSD_monitor(
COMPONENT nD_Mantid_1 = Monitor_nD(
1 COMPONENT nD_Mantid_1 = Monitor_nD(
2   options = "mantid square x limits=[-0.3 0.3] bins=128
3   y limits=[-0.3 0.3] bins=128, neutron pixel min=0 t,
4   list all neutrons",
5   xmin = -0.3,
6   xmax = 0.3,
7   ymin = -0.3,
8   ymax = 0.3,
9   restore_neutron = 1,
10  filename = "bank01_events.dat")
   AT (0, 0, 3.2) RELATIVE sampleMantid
```

Line 10, Column 3 Tab Size: 4 Plain Text

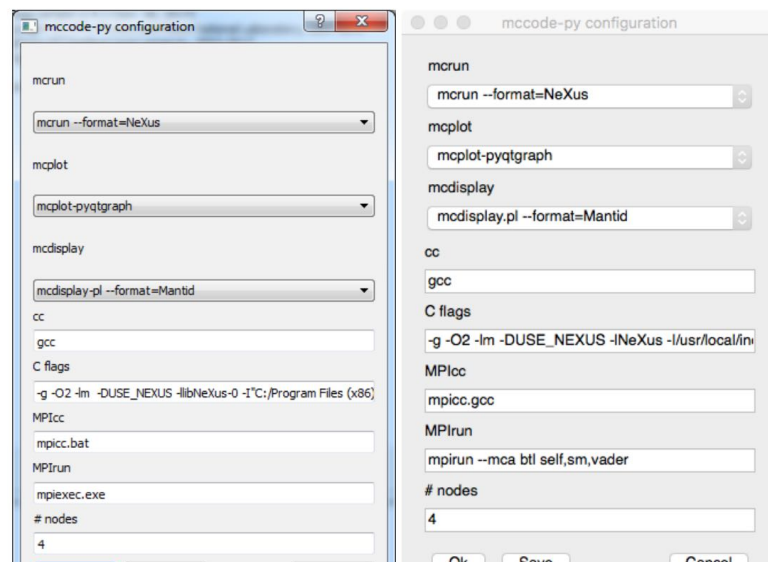
How to run the simulation



McStas GUI

Generating McStas event data for Mantid can be achieved from the McStas GUI `mcgui`. Below we show how to setup the simulation on Windows 7, OSX 10.12, and Ubuntu 16.04. For [McStas](#) we use version 2.4.1. For [Mantid](#) use version 3.4 or later.

1. Open the McStas configuration file. In `mcgui` go to: File -> Configuration
2. Change the setting as shown in figures below:
 - In the section `mcrun` select `mcrun --format=NeXus`
 - In the section `mcplot` select `mcplot-pyqtgraph`
 - In the section `mcdisplay` select `mcdisplay-pl --format=Mantid` (Windows) or `mcdisplay.pl --format=Mantid` (OSX or Ubuntu)
 - In the section `C flags` select this line depending on your OS
 - Windows: `-g -O2 -lm -DUSE_NEXUS -llibNeXus-0 -I "C:/Program Files (x86)/NeXus Data Format/include/nexus" -L "C:/Program Files (x86)/NeXus Data Format/lib/nexus"`
 - OS X: `-g -O2 -lm -DUSE_NEXUS -lNeXus -I/usr/local/include/nexus`
 - Ubuntu: `-g -O2 -lm -DUSE_NEXUS -lNeXus`



Semi-outdated
Wikipage on GitHub



Requires a serious rewrite!

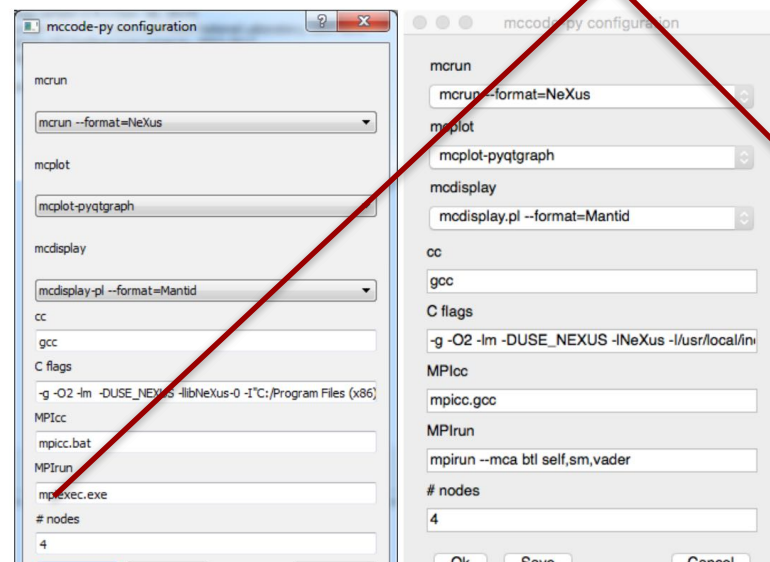
How to run the simulation



McStas GUI

Generating McStas event data for Mantid can be achieved from the McStas GUI `mcgui`. Below we show how to setup the simulation on Windows 7, OSX 10.12, and Ubuntu 16.04. For [McStas](#) we use version 2.4.1. For [Mantid](#) use version 3.4 or later.

1. Open the McStas configuration file. In `mcgui` go to: File -> Configuration
2. Change the setting as shown in figures below:
 - In the section `mcrun` select `mcrun --format=NeXus`
 - In the section `mcplot` select `mcplot-pyqtgraph`
 - In the section `mcdisplay` select `mcdisplay-pl --format=Mantid` (Windows) or `mcdisplay.pl --format=Mantid` (OSX or Ubuntu)
 - In the section `C flags` select this line depending on your OS
 - Windows: `-g -O2 -lm -DUSE_NEXUS -lNeXus -I "C:/Program Files (x86)/NeXus Data Format/include/nexus" -L "C:/Program Files (x86)/NeXus Data Format/lib/nexus"`
 - OS X: `-g -O2 -lm -DUSE_NEXUS -lNeXus -I /usr/local/include/nexus`
 - Ubuntu: `-g -O2 -lm -DUSE_NEXUS -lNeXus`



Semi-outdated
Wikipedia on GitHub



Requires a serious rewrite!

Instead, simply:

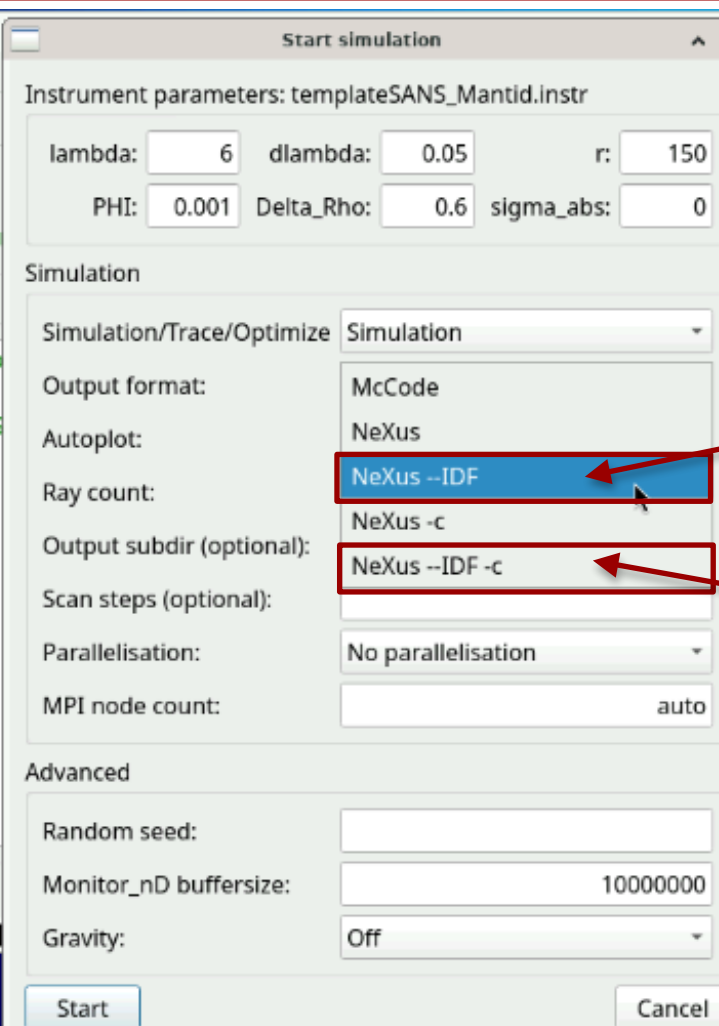
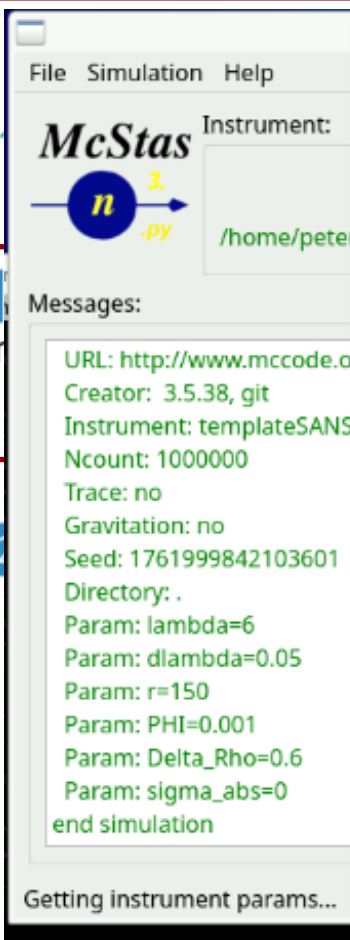
(In fact optional, but recommended...)

```
44 DEFINE INSTRUMENT templateSANS_Mantid(lambda=6, dlambd=0.05, r=150, PHI=1e-3, Delta_Rho=0.6, sigma_abs=0.0)
45 DEPENDENCY " @NEXUSFLAGS@ "
```

```
61 COMPONENT sourceMantid = Source_simple(
62   radius = 0.02, dist = 3, focus_xw = 0.01, focus_yh =
63   lambda0 = lambda, dlambd = dlambd, flux = 1e
64   AT (0, 0, 0) RELATIVE arm
```

```
81 SPLIT COMPONENT sampleMantid = Sans_spheres(
82   R=r, Phi=PHI, Delta_rho=Delta_Rho, sigma_abs=sig
83   xwidth=0.01, yheight=0.01, zdepth=0.005)
84 AT (0, 0, 3.2) RELATIVE coll2
```

```
106 COMPONENT nD_Mantid_1 = Monitor_nD(
107   options ="mantid square x limits=[-0.3 0.3] bins=1
108   xmin = -0.3,
109   xmax = 0.3,
110   ymin = -0.3,
111   ymax = 0.3,
112   restore_neutron = 1,
113   filename = "bank01_events.dat")
114 AT (0, 0, 3.2) RELATIVE sampleMantid
```

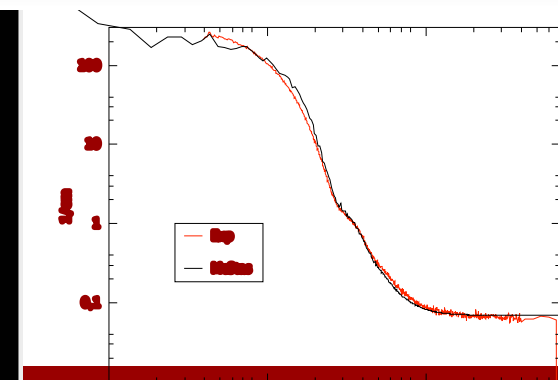
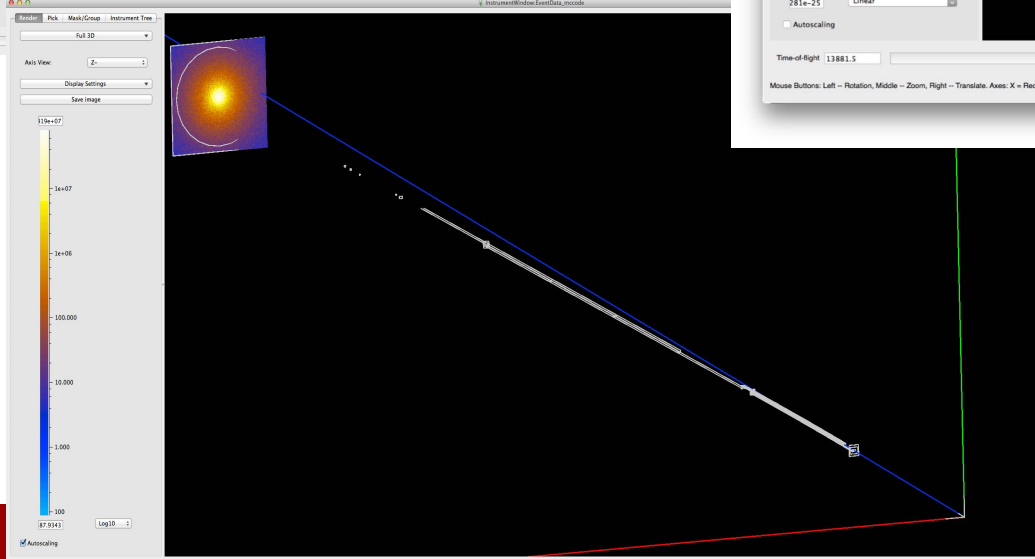
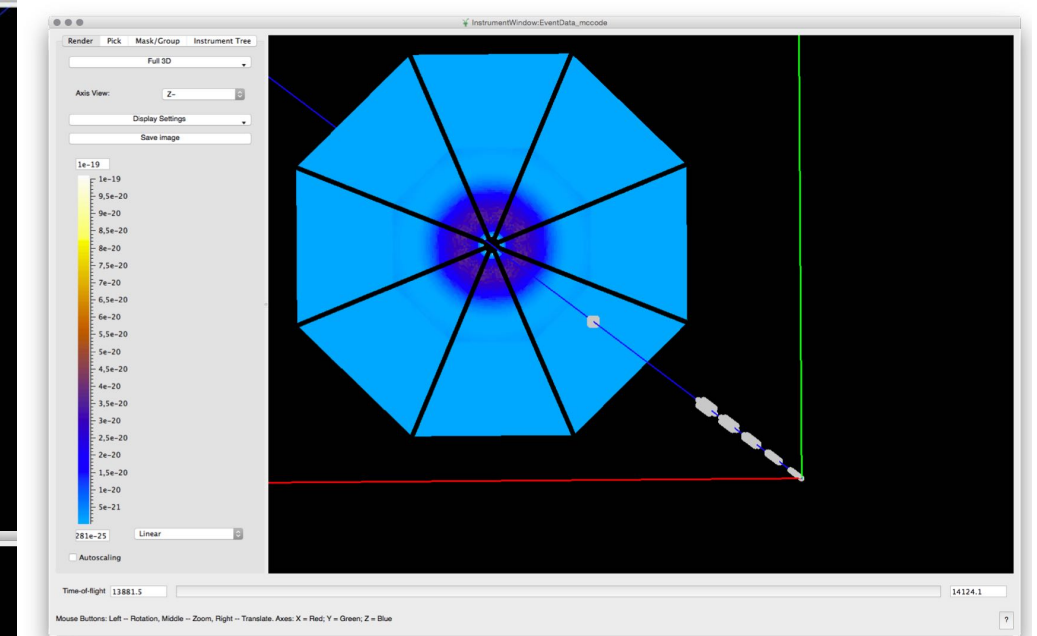
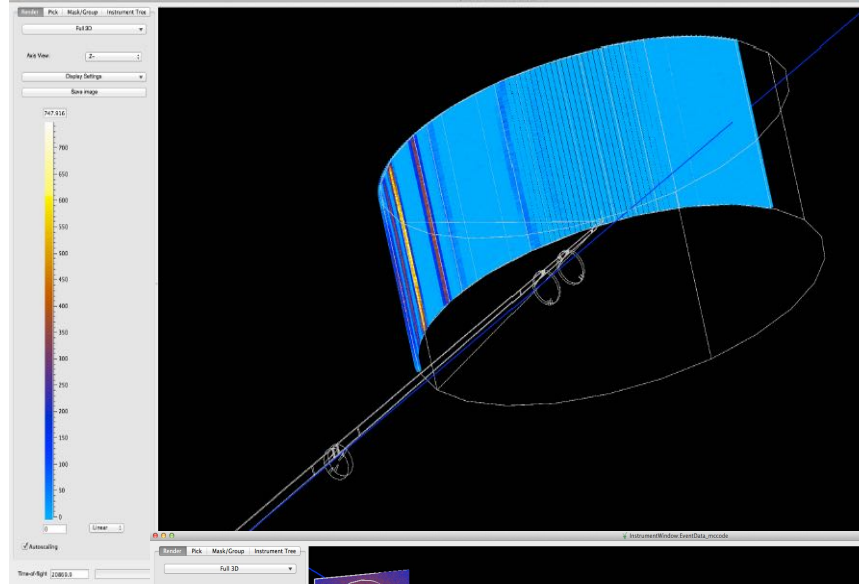


Request NeXus + IDF

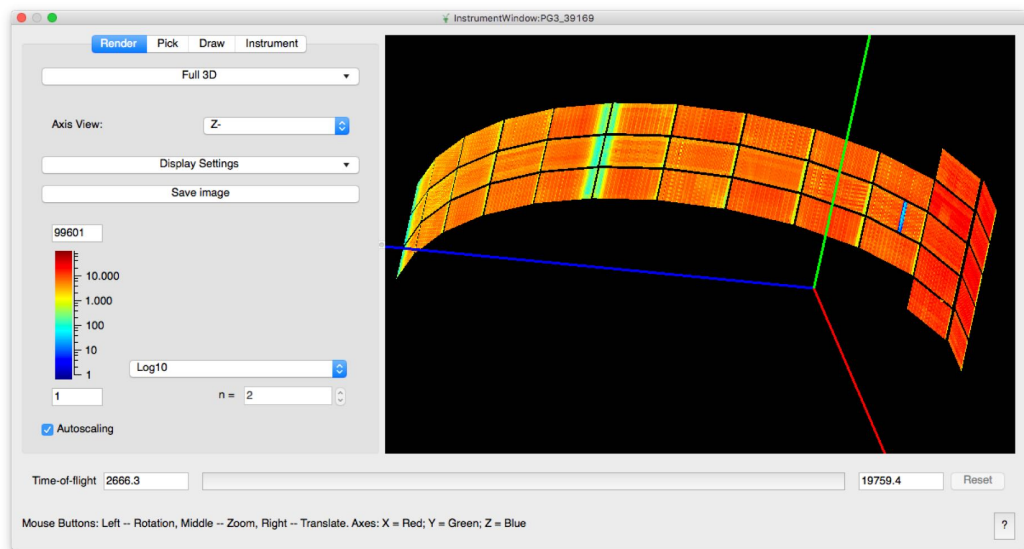
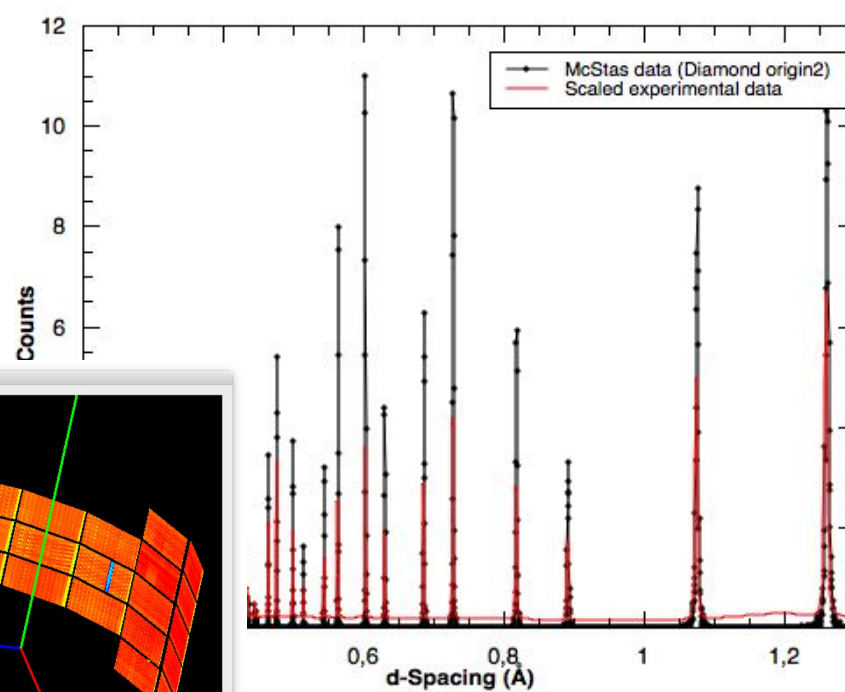
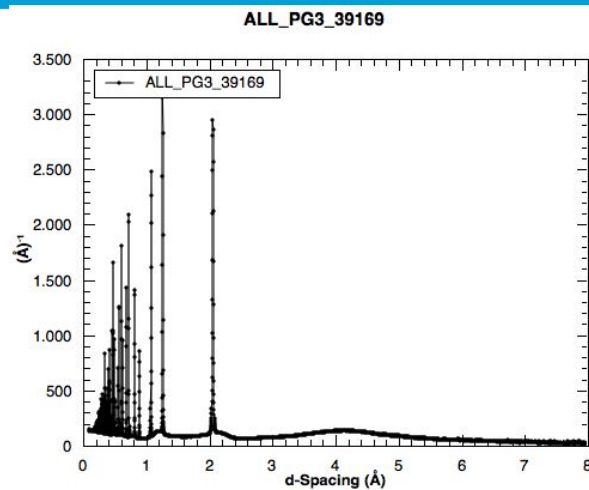
- same but do a recompile first



Examples of detectors: IN5 (ILL), LoKI (ESS), SANS2D (ISIS)

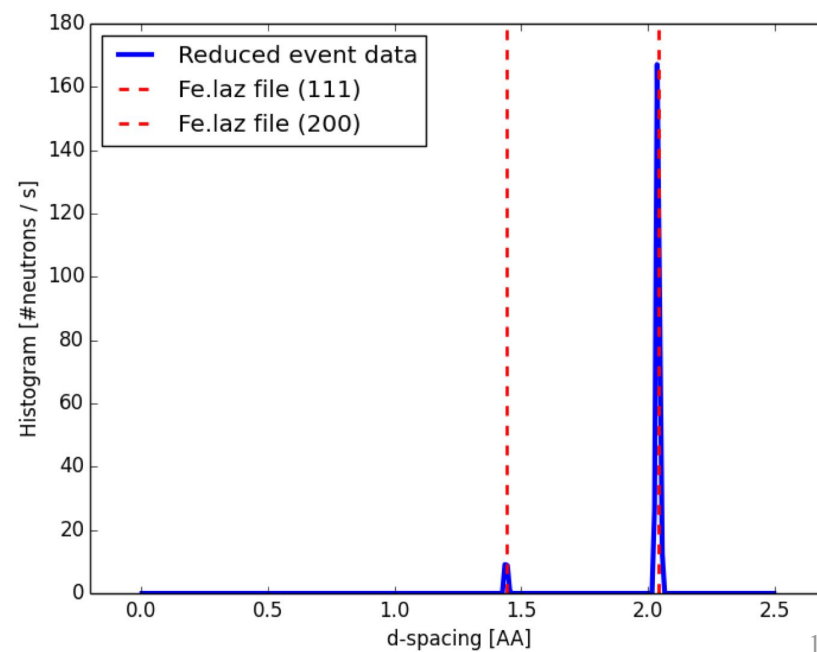
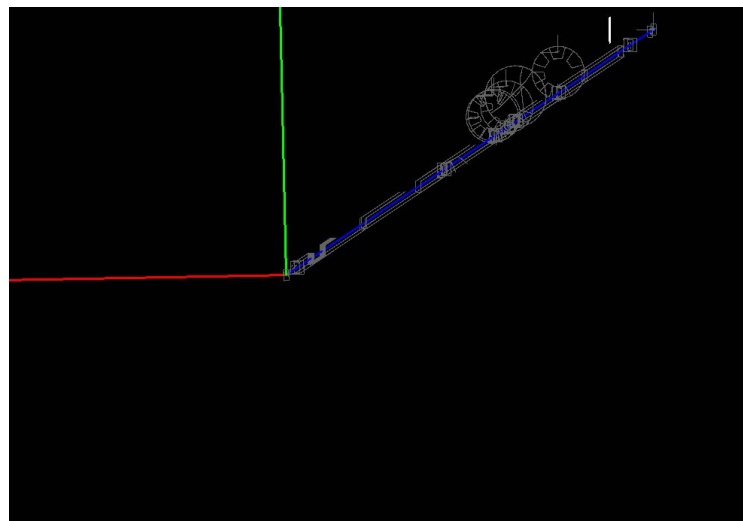


McStas and experimental data: POWGEN



By CelineD@ ESS + SNS collaboration

Example –V20 –Powder diffraction Fe





Summary

1. Easier to post-process (event) data than making a completely new McStas simulation / component
 - E.g.:
 - Rebin is needed
 - Sum selective over detectors
 - Use already developed methods in Mantid
2. Mantid process McStas data as if they were equal to experimental twin data



McStasToX - new kid on the block

1. Similar in spirit: Embed information to define k_i and k_f in the NeXus-file
2. 'X' is for now the scipp framework from ESS, but we expect further backends in the future, e.g. Mantid
3. Like scipp, plopp etc. from ESS can be used from a Jupyter
4. Repo at <https://github.com/mccode-dev/McStasToX/>
5. pip package only for now, conda on the way
6. Currently in beta-status but quite functional (see demo)



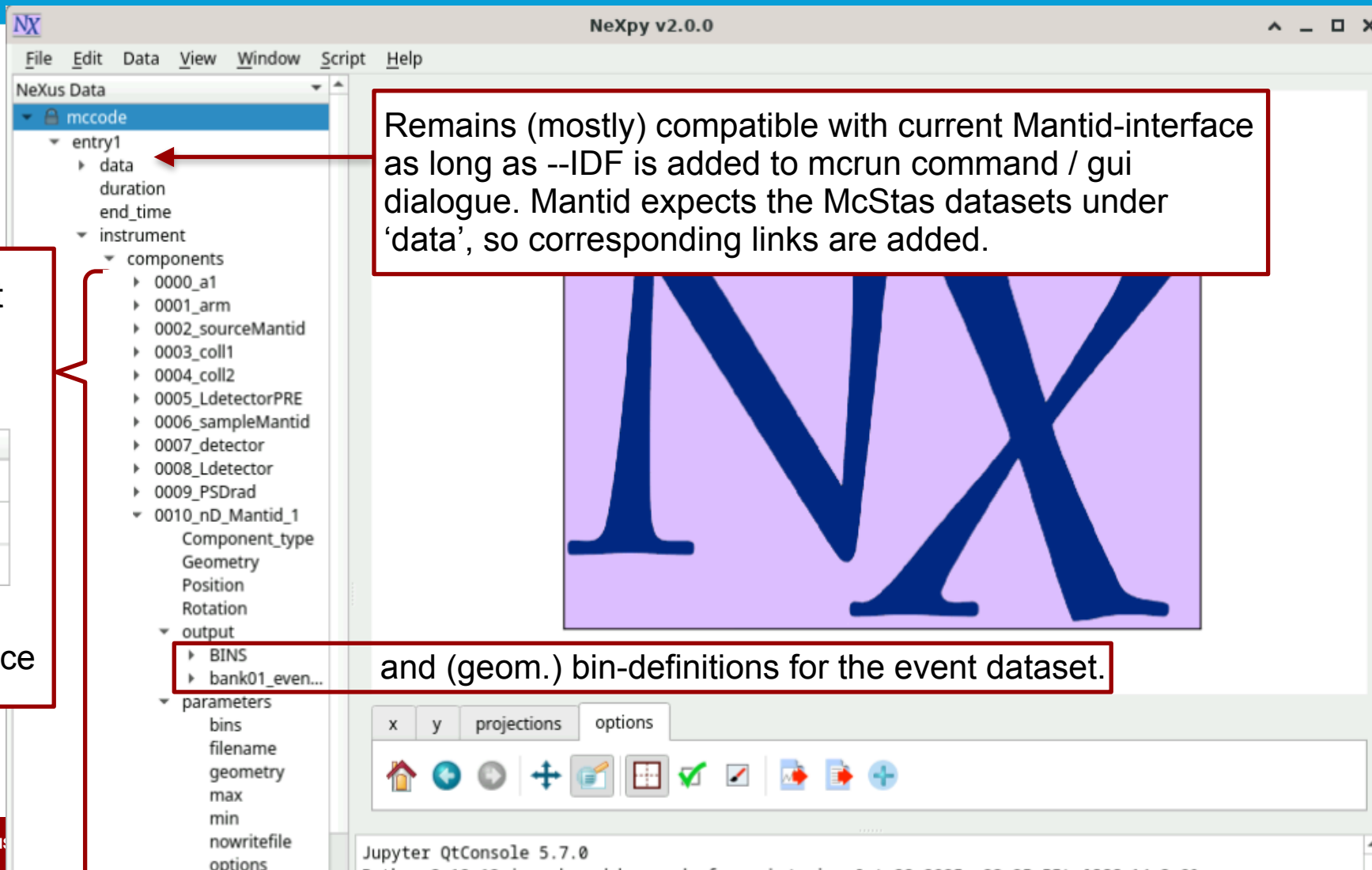
Full list of component data structures including pos and rot as datasets

	0	0	1	2
0 0	0 1	0	0	
1 0	1 0	1	0	
2 9.4	2 0	0	1	

as well as all params from the comp instance



McStasToX - new kid on the block, richer NeXus file content:



Remains (mostly) compatible with current Mantid-interface as long as --IDF is added to mcrun command / gui dialogue. Mantid expects the McStas datasets under 'data', so corresponding links are added.

and (geom.) bin-definitions for the event dataset.